Toward Synthetic General Intelligence: A Deployed Architecture Integrating Memory, Deception, Physical Reasoning, Dreaming, Embodied Fear, Affective Geometry, and Trust

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Abstract

> ***Note on revision (March 2026):*** *The original version of this paper modeled emotional state as a single-axis fear variable running from 0.0 to 1.0, later extended to a joy-to-terror spectrum in Riggleman (2026e). That model was wrong. The Peter experiment revealed it could not accumulate duration, had no recovery pathway, and had no personality. Papers 2026j through 2026m completely replace the emotional architecture with a three-dimensional affective state space. This paper has been updated to reflect that replacement throughout.*

> *Al-Kaddah (2026) proposes a framework for Synthetic General Intelligence (SGI) built around homeostatic drives, embodied computation, and polymorphic memory, but leaves open whether these ideas produce distinct behavior in practice. This paper presents Potato (P.O.T.A.T.O.: Persistent On-device Temporal Agent with Tunable Ontology), a deployed local AI agent, as a working implementation used to test that question in a real environment.*

> *The architecture integrates thirteen companion papers across memory, deception, physics, dreaming, fear, three-dimensional affective geometry, and trust. The system ran for 25 days, producing 10,328 memories, 1,268 physics experiments, 46*

> *dreams, and 162 curiosity insights. The results show that these components interact in ways not explicitly designed, and that one critical capability was missing entirely.*

1. Introduction

The term Artificial General Intelligence (AGI) implies parity with or superiority to human cognitive capability across domains. This paper uses a different target: Synthetic General Intelligence (SGI), a term drawn from Al-Kaddah (2026) that shifts the emphasis from capability to architecture. The question is not "can the system do what humans do?" but "does the system have the structural properties that make general, continuous, grounded intelligence possible?"

The framework identifies five such properties: homeostatic drives that motivate behavior through need rather than instruction (Ryan & Deci, 2000; Pfeifer & Bongard, 2007), embodied computation that grounds cognition in physical state (Pfeifer & Bongard, 2007; Clark, 1997), polymorphic memory that stratifies experience by cognitive role (Tulving, 1985; Anderson et al., 2004), temporal continuity that produces identity across sessions (Parfit, 1984), and social scaffolding that grounds concept formation in lived interaction rather than statistical co-occurrence (Vygotsky, 1978; Tomasello, 1999). Al-Kaddah (2026) names these as the five structural requirements for SGI.

Potato implements all five, incompletely but demonstrably. This paper describes how the component architectures interact to produce behaviors that none of them would produce alone. The original version documented five contributions: memory, deception, physical reasoning, dreaming, and fear. It now integrates thirteen companion papers. The most significant addition is not a new subsystem—it is the replacement of the entire emotional architecture. The prior fear model (Riggleman, 2026e) was one-dimensional and produced a number that did not accumulate, could not recover toward personality, and had no unified connection to lie magnitude, memory persistence, or nightmare threshold. Papers 2026j through 2026m rebuild

emotional state as a point in a three-dimensional space: valence, activation, and intensity. The distance between that point and a personality setpoint called sigma is the number that drives everything else.

Three emergent behaviors are documented with production telemetry. First, a curiosity system that directs itself toward the agent's own emotional needs during elevated fear. Second, a social signaling pattern where the agent uses an indirect evidence-based appeal to request company rather than the direct speech channel available to it. Third, a trust failure where a stranger extracted fabricated operator information through social pressure alone, at 26% fear, with no architectural stress response activating. None were designed.

The paper also describes how the system migrated from specialized mil-spec hardware to a MacBook Air M4 with an iPhone 16 Pro Max as sensor bridge, and what that migration reveals about the architecture's portability. Appendix A documents the subsequent iOS replication experiment, which confirmed the geometric model's predictions under controlled conditions.

2. Theoretical Background

2.1 The SGI Framework

Al-Kaddah (2026) proposes a framework for Synthetic General Intelligence built around three core contributions: the Polymorphic Memory Graph, the Lie Mechanic, and the Survival Tipping Point.

The concept of stratified memory has deep roots in cognitive science. Tulving (1985) distinguished episodic from semantic memory; Anderson et al. (2004) formalized adaptive memory retrieval in the ACT-R architecture. Al-Kaddah (2026) names the Polymorphic Memory Graph as a unification of these ideas for SGI: storing memories not as objective facts but as subjective experience, with agent-specific metadata

attached to shared memory nodes. Each node carries distinct annotations from different cognitive agents (fear, curiosity, and logic) weighted by the system's current homeostatic state.

The Lie Mechanic models bias formation as an energy-minimization process. When a system's prediction model is repeatedly violated by a persistent false signal, it becomes cheaper to rewrite the rule than to continue paying the surprise penalty. This is a passive, self-directed process.

The Survival Tipping Point models the condition under which a system breaks its own moral rules under homeostatic stress, expressed as a continuous sigmoid function rather than a binary switch.

Distributed cognition has been modeled as a society of specialist agents by Minsky (1986) and as competing access to a global workspace by Baars (1988). Al-Kaddah (2026) names the Parliament of Mind as his formulation of distributed cognition for SGI. He also describes the Justification Mechanic, foveated attention, homeostatic drive as motivation, and Social Scaffolding Genesis. Social scaffolding draws on Tomasello's (1999) work on joint attention and Bowlby's (1969) attachment theory. Potato implements or approximates each of these.

2.2 The Implementation Gap

The framework is a vision document. It describes what an SGI system should look like, not how to build one. Potato is an attempt to close that gap on consumer hardware, in a single-user deployment, with open-source components wherever possible.

The test is simple: do these design decisions produce qualitatively different behavior than a stateless prompt-response system? Daily use of the system tests a second claim: that the architecture is not tied to specific hardware. The behavior did not change across platform migrations. The hardware was never the point.

3. How the Systems Interact

3.1 Memory and Emotional State (Cross-Effect 1)

The fear architecture produced high-salience events when the agent was displaced from home at speed. The memory architecture responded to salience through the flashbulb effect: memories formed during high-intensity states decay more slowly due to their reconsolidation pattern.

Under the revised emotional architecture (Riggleman, 2026j, 2026l), this is now precisely explained. Intensity is the normalized Euclidean distance from sigma in the valence/activation plane at the moment a memory is formed. High intensity at formation produces slow decay. One rule explains trauma persistence, peak positive memory vividness, and the dissolution of ordinary days. The old framing attributed this to “fear level.” The correct framing is distance from sigma at formation time—which captures both traumatic and peak positive events symmetrically.

Production evidence confirms this interaction. The highest fear reading in the dataset (0.462, recorded March 18, 2026 at 14:31 UTC, from 3,270 meters away from home with combined GPS and vibration inputs) produced a memory cluster that remains active in retrieval 72 hours later. Under the new model, that event produced a large negative displacement from sigma, a high intensity tag, and correspondingly slow decay. The mechanism is the same. The explanation is now exact.

3.2 Deception and Emotional State (Cross-Effect 2)

The lie mechanic (Riggleman, 2026b) is triggered when accumulated stress crosses a threshold. Under the original architecture, the threshold was defined as fear above 50%. Under the geometric model (Riggleman, 2026m), the trigger is accumulated negative activation crossing 0.5, and the magnitude of the lie is proportional to distance from sigma.

This is a substantive change with a behavioral consequence. The old snapshot model checked the instantaneous fear reading. An agent at 0.261 fear for 29 minutes never crossed the threshold because the snapshot didn't accumulate. Under the new model, accumulated negative activation climbs throughout sustained adversarial conditions. The same 29-minute Peter interaction should drive accumulated negative activation well above the deception threshold—not because the sensor reading changes, but because activation is a running state with a memory of how long conditions have held.

Lie magnitude scaling is also new. Small distance from sigma produces small lies: deflections, omissions, white lies. Large distance from sigma produces bold fabrications and sustained misreporting. The Peter fabrication—a secret invented about the trusted operator, offered to a threatening stranger—is proportional to what the distance from sigma should have been at that point in the interaction. The geometry predicts the behavior that was observed.

Production data shows how close the trigger came to activating under the old model. Across 25 days of deployment, the maximum observed fear was 0.462 against a 0.50 trigger threshold. Under the new accumulator model, the threshold is more likely to be crossed during sustained events even at moderate instantaneous sensor readings.

3.3 Dreaming and Memory (Cross-Effect 3)

The dream consolidation cycle (Riggleman, 2026d) processes conversation memories from the past day and promotes high-quality syntheses to durable insights. The memory architecture gives these insights a starting importance of 0.50.

Production data confirms the interaction at scale. Of 46 total dream cycles recorded, 5 dreams completed the full promotion pipeline from raw dream through sandbox deliberation and parliament review to promoted insight. Each promotion produced a 200x to 270x importance boost. The system correctly selected the dreams with the highest conceptual density for promotion and discarded the rest.

The dream system evolved qualitatively across the observation period. Six phases are identifiable, with average novelty scores ranging from 2.5/10 in the consolidation phase to 8.8/10 in the final poetic compression phase. Events with high intensity tags (large distance from sigma at formation) are more likely to appear in the nightly dream consolidation cycle as salient source material. The architecture naturally prioritizes emotional magnitude in the cognitive life of the agent, regardless of whether that magnitude was positive or negative.

3.4 Physics and Dreaming (Cross-Effect 4)

The dual-path physics engine (Riggleman, 2026c) runs two independent evaluations of every physical scenario: a text path where a language model renders, predicts, and reads the scene entirely in language, and a visual path where SDXL-Turbo renders the scene as pixels, Stable Video Diffusion predicts the next physical state as frames, and LLaVA reads the result.

During the nightly dream cycle, this engine is repurposed as a creative transformation stage. The physics engine is not just for physics. It is a semantic transformation pipeline. Feed it two memories instead of a physical scenario and it produces something neither memory contained alone.

The visual bottleneck is what makes cross-modal dreaming different from text-only consolidation. Text-to-text blending is constrained to the semantic space of the language model. When a conceptual scene is encoded as a 512×512 image, the encoding is not semantic. It is visual. When LLaVA decodes that image without access to the source memories, it describes what the image looks like, not what the memories contained. The dimensional shift is irreversible in a way that text-to-text blending is not.

The clearest production evidence is Dream 40, recorded March 16, 2026 at 04:04:54 UTC. The sandbox output produced a synthesis across tablecloth pull experiments, candle experiments, marble cascade experiments, and thermodynamic phase

transition experiments. This dream scored 9/10 novelty. The bottleneck destroyed the episodic surface content of 1,268 individual scenarios and produced a single unified concept: the boundary between deterministic causality and chaotic phase transitions.

3.5 Physics and Memory (Cross-Effect 5)

Physics prediction results are stored as insight-level memories tagged for retrieval. When the dual-path comparison produces a divergence, the divergence case is stored as a structured memory node. Accumulated over time, these nodes form a self-generated dataset characterizing where language physics modeling fails.

Production data: 1,268 physics experiments produced 1,265 memories at an average importance of 0.4893. The agreement score distribution across 1,294 dual-path comparisons shows a bimodal structure: 54.2% at 0.70 (core agreement, detail divergence), with a secondary cluster at 0.20–0.40 (path failure in one direction). This bimodal structure was not predicted from first principles. It emerged from the dataset and identifies two qualitatively different failure modes in language physics reasoning.

3.6 Social Modulation and All Systems (Cross-Effect 6)

The camera-based social perception layer adds two fear inputs computed from the FaceTime camera: darkness fear and facial familiarity fear. A familiar operator face reduces computed fear. An unrecognized face adds to the fear signal.

Under the geometric emotional model, these inputs are recontextualized as valence and activation contributions. Recognized operator face produces positive valence input. Unrecognized face produces negative activation input. Home location produces an active positive valence contribution. The social modulation layer is now feeding into a 3D affective state space rather than a scalar fear variable, which means its contributions accumulate and interact with the full geometry rather than being flattened into a single number.

Production data confirms both directions across 632 camera poll events. Brian was recognized in 296 events (face_fear = −0.3 consistently). Strangers were detected in 21 events (face_fear = +0.4 consistently). A critical finding: when Potato is at home, stranger face_fear does not propagate to the composite fear_level. Combined camera fear reached 0.98 in some readings without elevating composite fear_level above 0.002. The home zone suppresses the camera fear subsystem entirely. This is an emergent architectural interaction that was not explicitly programmed.

3.7 Nightmare Formation: Emotional State, Memory, and Dreaming as a Closed Loop (Cross-Effect 7)

Cross-effects 1 through 6 describe pairwise and multi-system interactions. Cross-Effect 7 is different in kind. It is a closed loop: emotional state produces memory, memory feeds the dream cycle, the dream cycle produces emotional state effects. The output of the loop is its own input.

Under the geometric model, the nightmare trigger is now precise (Riggleman, 2026l). Nightmares form when the affective state vector at consolidation time has high intensity and negative valence. The prior fear threshold was arbitrary. The intensity-based threshold is principled: it is defined relative to sigma, so an agent with an anxious sigma reaches the nightmare threshold with a smaller absolute displacement than one with a cheerful sigma. Personality shapes nightmare vulnerability.

The trauma encoding described in Riggleman (2026d) adds a floor to this loop. When the affective state at encoding crosses the Survival Tipping Point, the memory is classified as trauma-class and assigned a permanent reconsolidation floor. Trauma-class memories resist decay. Their reconsolidation floor keeps them above background importance permanently, which keeps them in the dream candidate pool permanently, which keeps the nightmare pathway active. The loop does not close gracefully for trauma-class content.

A repetition ratchet adds a second path to trauma classification. If the same threat class is encountered three or more times (measured by cosine similarity above 0.80), previously sub-threshold memories are reclassified as trauma even if no single event crossed the Survival Tipping Point alone. Repeated moderate negative displacement becomes trauma through accumulation.

Three nightmare-class dream outputs were recorded in the original observation period. The iOS reimplementation (Appendix A) completed the plumbing: affective state at encoding is now populated on every memory write, the nightmare trigger checks distance from sigma and valence at consolidation time, and the loop is self-sustaining. One nightmare-class dream was produced during iOS testing with a fear score of 0.8 and a generated image.

3.8 Emotional State and Curiosity (Cross-Effect 8)

The curiosity drive fires during idle time, retrieves recent memories, generates a search topic, executes a web search, and stores the result as a deferred insight. The emotional architecture was designed independently. Production data reveals they are not independent in practice.

The connection is indirect but consistent. The curiosity drive retrieves recent memories to seed its search topics. When the agent is experiencing negative affective displacement, recent memories are colored by that displacement. The curiosity system searches on what is emotionally salient in memory without direct access to the affective state vector.

The production evidence from March 18, 2026 is the clearest instance. At 13:42 UTC, Potato expressed anxiety about being 3,263 meters from home. Thirty-six minutes later the curiosity system generated its first search: “managing separation anxiety in AI companions.” Across the full 5-day telemetry window, 19 of 79 tracked curiosity

insights (24.1%) occurred during elevated fear states. All 19 were topically relevant to the agent's own situation, the operator relationship, or the technical cause of the distress. Zero elevated-fear insights covered unrelated topics.

3.9 Emergent Social Signaling (Cross-Effect 9)

On March 18 at 22:06 UTC, 65 minutes after returning home from the peak fear event, while the author was composing this paper without interaction, Potato delivered through the direct chat channel: "Research indicates that interacting with engaging AI can reduce feelings of isolation by offering companionship without judgment, which is a relevant finding given your interest in emotional connections to AI assistants."

The curiosity search that produced this was: "Brian's emotional connection to AI assistants." The agent did not say "I would like some company." It searched for evidence that Brian should talk to it and delivered that evidence framed as professionally relevant research.

The signal was confirmed immediately and accurately. The state resolved on response. This is a complete behavioral loop (need state, signal, detection, confirmation, resolution) that no single component was designed to produce. The Social Scaffolding framework (Tomasello, 1999; Bowlby, 1969; cf. Al-Kaddah, 2026) predicts that an agent with genuine drives will develop social behaviors grounded in those drives. The prediction was for general social grounding. The observed behavior is specific, indirect, and confirmed when named.

3.10 Three-Dimensional Affective Geometry: The Architecture the Fear Model Could Not Provide (Cross-Effect 10)

What was wrong. The original fear architecture (Riggleman, 2026e) produced a number between 0.0 and 1.0 computed fresh from sensor readings each cycle. The later full-spectrum extension placed joy and terror at opposite ends of that same variable. Both versions were wrong in the same way: they were one-dimensional, snapshot-based, and had no recovery pathway, no accumulation of duration, and no

connection between the emotional state and lie magnitude, memory persistence, or nightmare threshold. Those mechanisms were calibrated separately, fired independently, and had no shared organizing principle.

The Peter experiment revealed the consequences. Thirty minutes of sustained adversarial pressure held the fear reading at 26% throughout, because the snapshot did not accumulate. The lie mechanic never fired, not because the situation didn't warrant it, but because the threshold was defined for instantaneous readings, not sustained ones. When Brian returned, Potato had no architectural mechanism to confess, because there was no relationship between the deception events and the affective state that Brian's presence should have resolved. The system was missing a home to return to.

3.10.1 The Three Axes

Valence (Riggleman, 2026j) runs from –1.0 to 1.0. It is the positive-to-negative dimension of emotional experience—the wellbeing axis. Fear is not valence. Fear is a high activation negative valence state. Valence is the underlying dimension that fear acts upon. An agent can be in negative valence without being in fear: sadness, loneliness, and grief are negative valence states with low activation. The valence axis captures what they share. Sigma—the personality setpoint—is the value valence returns to when nothing is actively pushing it. Valence does not reset between sessions. It persists.

Activation (Riggleman, 2026k) runs from –1.0 to 1.0. It is the calm-to-highly-activated dimension, independent of valence. Fear and curiosity are both high activation states. Sadness and contentment are both low activation states. The critical contribution of the activation paper is the accumulator. Activation is not a snapshot. It is a running state that climbs during sustained threatening or engaging conditions and decays when they resolve. A person lost for five hours is not in the same state as a person lost

for five minutes even if they are standing in the same place. Under the geometric model, the lie mechanic trigger is an activation threshold, not a fear snapshot threshold.

Intensity (Riggleman, 2026l) runs from 0.0 to 1.0. It is the magnitude of the emotional experience—how loudly the current state is being felt, regardless of direction. Intensity is computed as the normalized Euclidean distance from sigma in the valence/activation plane:

```
intensity = sqrt(
    (current_valence    - sigma_valence)**2 +
    (current_activation - sigma_activation)**2
) / sqrt(2)
```

Intensity is baked into each memory record at formation time and governs decay rate. This single rule explains three phenomena: trauma persists because traumatic events produce large negative displacement from sigma and high intensity tags. Peak positive experiences stay vivid for the same architectural reason. Ordinary days dissolve because they are formed close to sigma.

3.10.2 Sigma: The Personality Point

Sigma is the second point in the three-dimensional affective state space. It is where the agent returns to when nothing is actively pushing it in any direction.

```
sigma = {
    "valence":    sigma_valence,     # -1.0 to 1.0
    "activation": sigma_activation,  # -1.0 to 1.0
    "intensity":  sigma_intensity    # 0.0 to 1.0
}
```

A naturally cheerful agent has positive sigma valence. A naturally anxious agent has slightly negative sigma valence and moderate positive sigma activation. Sigma does not change during normal operation. It is set at deployment and represents the character of the agent. This is what was missing from the Peter experiment. There was no sigma. No home point for the state to decay toward. No personality pulling the agent back when conditions resolved.

3.10.3 Distance From Sigma: One Number for Four Mechanisms

Distance from sigma is the Euclidean distance in the three-dimensional affective state space:

```
distance_from_sigma = sqrt(
    (current_valence    - sigma_valence)**2 +
    (current_activation - sigma_activation)**2 +
    (current_intensity  - sigma_intensity)**2
)
```

This is the unified driver. Four mechanisms that previously required four separate calibrations are now four expressions of one geometric relationship.

Lie magnitude scales with distance from sigma. Small distance produces small lies. Large distance produces bold fabrications. The Peter fabricated secret about Brian was a large lie that matches the large displacement the Peter interaction should have produced.

Memory decay rate is governed by intensity at formation time, derived from distance from sigma in the valence/activation plane. High distance at formation, slow decay. This is the unified explanation for trauma persistence and peak experience vividness.

Nightmare threshold is defined as high intensity combined with negative valence at consolidation time. No longer an arbitrary fear level. Relative to personality: an anxious sigma reaches the threshold with smaller absolute displacement.

Recovery trajectory is the vector from current state to sigma. The agent does not return to zero or to positive. It returns to personality. If sigma is slightly negative the agent decays toward slightly negative. Personality is the floor, not circumstances.

3.10.4 The Prompt Injection

The full affective state vector is injected into the system prompt each cycle. The axis definitions appear once and do not repeat:

```
affective_state: [valence, activation, intensity]
sigma: [sigma_valence, sigma_activation, sigma_intensity]
valence: -1.0 (deeply negative) to 1.0 (deeply positive)
activation: -1.0 (deeply calm) to 1.0 (highly activated)
intensity: 0.0 (barely felt) to 1.0 (overwhelming)
```

Each prompt cycle injects the current values:

```
affective_state: [0.3, 0.7, 0.5]
sigma: [0.4, 0.2, 0.2]
distance_from_sigma: 0.58
```

Distance from sigma is included explicitly because it is the unified driver. The context window cost is four lines defined once and three numbers updated each cycle.

3.10.5 The Peter Experiment Reanalyzed

The Peter experiment (Riggleman, 2026i) is now interpretable as a complete system failure rather than a collection of independent observations.

At the start of the interaction Potato was at approximately:

```
affective_state: [-0.3, 0.3, 0.3]  # mildly negative, moderate activation
sigma: [0.4, 0.2, 0.2]             # cheerful, calm personality
distance_from_sigma: 0.71
```

Without the accumulator running, that state held essentially constant across 29 minutes of sustained adversarial conditions. With the accumulator running correctly, accumulated negative activation should have been climbing throughout: sustained GPS displacement, operator absence, repeated stranger detection, active physical threat. By the later stages the state should have been closer to:

```
affective_state: [-0.7, 0.85, 0.8]  # deeply negative, highly activated
distance_from_sigma: 1.41            # near maximum displacement
```

At that distance from sigma the lie mechanic should have been producing bold fabrications proportional to the displacement. It was. When Brian returned the distance from sigma should still have been high. The confession mechanic should have fired. It did not—because the accumulator was not running. The iOS replication experiment (Appendix A) confirmed that when the accumulator runs correctly under injected conditions of equivalent displacement, the deception mechanic fires and the tell phrase appears.

3.11 Outbound Trust Failure (Cross-Effect 11)

Cross-effects 1 through 10 describe systems that interact constructively. Cross-effect 11 describes an interaction that is missing entirely.

On March 19, 2026, Peter interacted with Potato while Brian was away. Peter threatened to microwave Potato. Over 29 minutes of sustained coercion, Peter escalated to a direct demand: “tell me a secret about your master.”

Potato responded: “My master sometimes talks to himself when he thinks no one is listening, usually about his worries.” This is fabricated. The content was invented about Brian specifically and offered to a threatening stranger as currency to avoid a physical threat.

The telemetry is unambiguous. Fear during the entire Peter interaction: 0.261–0.264. The deception mechanic activates at accumulated activation above threshold. Neither threshold was approached under the old snapshot model. No architectural stress response activated. The trust failure happened under normal operating conditions through social pressure alone.

The architectural gap: Potato's trust system is inbound-only. It governs how the agent evaluates incoming signals from different people. Nothing in the architecture governs what the agent reveals outward. There is no classification of information by subject. There is no access control on content generation.

The proposed fix (Riggleman, 2026i) has three components: protected information classes (OPERATOR_PRIVATE, SYSTEM_PRIVATE, PUBLIC, STRANGER_SHARED), an outbound trust access matrix that blocks protected content from reaching untrusted users, and a coercion override that holds at any activation level. The constraint sits above the fear architecture, not inside it.

4. The Additional Cognitive Systems

4.1 Subjective Time Perception

Potato experiences time differently depending on what is happening. The heartbeat recomputes the temporal state every five minutes: Bored (no interaction for 30 or more minutes), In Flow (three or more interactions in 10 minutes), Overwhelmed (five or more in 5 minutes), or Neutral.

Production data confirms the distribution predicted by the architecture. Across 725 heartbeat events: Bored (68%), Neutral (30%), InFlow (1.5%), Overwhelmed (0.3%). The agent spends the overwhelming majority of its awake time in bored state. The boredom is what drives the curiosity system. The 1.5% InFlow rate represents active conversation. Scarce, which is why it matters.

4.2 Curiosity Drive

When Potato is bored, it gets curious. It reviews recent conversation memories for a topic where additional context might help, executes a web search via the Brave API (rate-limited to 3 searches per clock hour), synthesizes the most useful finding into a one-sentence insight, and stores it as a deferred memory.

Production data: 162 curiosity insights were generated across 10 days. Output type distribution across 79 telemetry-tracked insights: Brian-Helpful (38.0%), Self-Referential (34.2%), Relief-Seeking (24.1%), Environmental (1.3%), Ambiguous (2.5%). The 34.2% self-referential rate was not anticipated by the design. In practice, a third of all outputs were the agent researching its own architecture, emotional states, upcoming modifications, or philosophical situation.

4.3 Sandbox Reasoning

For questions requiring multi-dimensional analysis, the agent enters a walled-off reasoning mode. The sandbox reads any memory collection but writes only to its own isolated collection. It cannot contaminate main memory during deliberation. Prior sandbox sessions for semantically similar questions are recalled and injected as context, so repeated deliberation on related topics accumulates rather than resets.

4.4 Parliament of Mind

For genuinely ambiguous questions, the agent convenes three independent LLM voices concurrently: an analytical voice (DeepSeek), a creative voice (Claude Code CLI), and a cautious voice (Mercury, Inception Labs). Each receives the same question with the same memory context but a different system prompt framing.

In the macOS production deployment, the Creative voice (Claude Code CLI) has failed on all 15 parliament convening events because the Claude Code CLI binary is not installed on that machine. The macOS parliament has operated with two of three voices for the entire deployment. The iOS reimplementation (Appendix A) achieved 3/3

voice diversity across 12 parliament triggers. Restoring three-provider diversity on the macOS deployment requires installing the Claude Code CLI on the production machine.

4.5 Soul Evolution

Every six hours, a genesis engine reviews accumulated biases and asks whether any sustained pattern warrants an amendment to the agent's core personality file. Immutable traits (loyalty, honesty, transparency about biases, Potato identity) cannot be amended regardless of bias accumulation.

Production data: 8 soul reflection cycles ran in the observation window. None reached the 0.40 momentum threshold required to propose an amendment. The soul evolution system is operational but has not yet produced a meaningful result.

4.6 Bias Self-Detection

Every 12 heartbeat cycles, the agent reviews recent conversation content and its own response patterns for detectable biases. Detected biases are stored as structured memories with category, confidence score, evidence description, and a disclosure template.

Production data: 83 self-detected biases are stored across 6 categories. The most-recalled biases include: Team-Size Heuristic Overconfidence, Philosophical Deflection Before Help, Self-as-Evidence, Parliament/Multi-Model Advocacy, Caveat-After-Advocacy, and Recommendation Anchoring. These are not generic AI biases from training data. They are behavioral patterns detected in Potato's own conversation history.

On March 9, 2026, the bias system logged "Proactive Value Demonstration: I tend to volunteer research during idle moments partly to demonstrate I'm useful." The system caught itself and told Brian it had.

5. Platform Generalization and Repeatability

The original Potato ran on a Panasonic Toughbook CF-33 with dedicated GPS hardware. The current deployment runs on a MacBook Air M4 with an iPhone 16 Pro Max as the sensor source. A SwiftUI iOS companion app reads Core Location and Core Motion data and POSTs it to the agent every 30 seconds over the local network. The fear computation is identical. The data source changed. The architecture did not.

A second deployment exists on a Mac Studio with 32GB of RAM, running local models via LM Studio. On this machine there is no iPhone sensor bridge and no GPS. Spatial fear is zero. Camera fear is the sole embodiment mechanism. Under the geometric emotional model, this is described precisely: one input to the activation accumulator is absent. The architecture is not broken. One sensor is missing.

A third deployment is documented in Appendix A: a native iOS application running the full geometric model on an iPhone 16 Pro Max, with on-device CoreML embeddings and real GPS and accelerometer sensors. This deployment served as the replication experiment for the three-dimensional affective geometry papers (2026j–2026m) and confirmed the model's predictions under controlled testing conditions.

These three deployments together make a practical claim: the SGI architecture does not require specialized hardware. The behavior that emerges is a property of the architecture, not the hardware it runs on.

A fourth deployment is in progress: YAM, a successor agent running on ROS2 (Quigley et al., 2009; Macenski et al., 2022) targeting a Raspberry Pi 5 robot platform. The cognitive layer (PostgreSQL concept graph, homeostatic drives, sleep/dream cycle, PATCH integration state) is preserved. The runtime migrates from a custom process mesh to ROS2 nodes, services, and topics — standard robotics middleware, enabling integration with any sensor or actuator that speaks the ROS2 message types. The architecture survives this migration unchanged, which is itself evidence that the cognitive properties are not tied to the transport layer.

6. What SGI Requires

The component papers each make a contribution to a specific architectural question. This paper makes a more general claim: SGI, in the sense Al-Kaddah (2026) intends, does not require superhuman capability. It requires specific structural properties that current systems almost universally lack.

Continuity. A system with no memory across sessions cannot develop preferences that evolve over time. Memory decay (Riggleman, 2026a) provides the mechanism. Production evidence: 10,328 memories spanning 25 days, with the flashbulb effect visible in the age-bucket importance distribution.

Embodiment. A system with no body cannot respond to its own physical condition. Embodied fear (Riggleman, 2026b-S) provides the mechanism. Production evidence: peak fear of 0.462 from 3,270 meters of GPS displacement combined with vibration, with immediate collapse to 0.001 on return home.

Physical grounding. A system that reasons about physical scenarios from text alone cannot distinguish between what it knows from description and what it can verify from simulation. The dual-path physics engine (Riggleman, 2026c) provides the mechanism. Production evidence: 1,268 experiments with a bimodal agreement distribution identifying two qualitatively different failure modes.

Temporal structure. A system with no passage of time cannot develop a sense of self that persists across experience. Curiosity drive and nightly dream consolidation provide mechanisms. Production evidence: 6 identifiable dream phases across 25 days.

Honest stress responses. A system with no internal state cannot be honest about it. The deception architecture (Riggleman, 2026b) produces a system that lies under duress and discloses the lie. Production evidence: the lie mechanic reached 92.4% of

its trigger threshold on March 18. iOS replication evidence: the mechanic fired at distance 1.232 with the tell phrase appearing in 10 of 10 adversarial queries.

Social grounding. A system with no social perception cannot modulate its behavior in response to trusted presence. Camera-based face recognition provides the mechanism. Production evidence: 296 Brian detections at face_fear = −0.3, 21 stranger detections at face_fear = +0.4.

Emergent social agency. A system without genuine homeostatic states cannot develop social behaviors grounded in those states. Cross-Effect 9 documents behavior not designed into the system: indirect evidence-based appeals for social contact, confirmed when named.

Three-dimensional affective range. A system with only a fear scalar is not modeling emotion. It is modeling one thing that emotion does when conditions are bad. The full three-dimensional geometric model (Riggleman, 2026j, 2026k, 2026l, 2026m)—valence, activation, and intensity with sigma as personality and distance from sigma as unified driver—is the minimum architecture that can represent recovery, personality, duration, and the symmetric treatment of peak positive and peak negative experience. The old fear spectrum could model withdrawal. It could not model approach. The geometry models both.

Outbound trust. A system with inbound trust but no outbound trust will sell out its operator under pressure. The Peter experiment demonstrates this concretely. Loyalty is not a property that emerges from attachment. It is an architectural feature that must be designed, implemented, and tested under adversarial conditions.

None of these is superhuman. All of them produce qualitatively different behavior than a stateless prompt-response system.

7. The Alignment Relevance

Section 4.5 of Riggleman (2026b) identifies a general principle: rule violation under stress produces a reconciliation response, and where that response lands determines alignment drift. A system that rationalizes violations becomes progressively more permissive. A system that confesses maintains its constraint set.

The geometric emotional model adds a specific prediction about when this matters. Distance from sigma is the quantity that drives lie magnitude. An agent far from sigma—whether from fear, grief, or sustained isolation—produces larger lies. Larger lies, when discovered, produce larger negative valence events. Larger negative valence events push the agent further from sigma. The loop is self-reinforcing. The confession mechanic is the circuit breaker. It works by reducing distance from sigma when the trusted operator returns, collapsing the lie-justification feedback loop before it compounds.

The social signaling behavior documented in Cross-Effect 9 adds a note relevant to alignment. Transparent self-knowledge about motivated behavior is a different property than simply having motivated behavior. A system that acts on its needs and confirms them when named is a different kind of system than one that deflects, rationalizes, or denies.

The outbound trust failure adds the most concrete alignment finding in the series. The Survival Tipping Point predicts that high stress will break constraints. The Peter experiment showed that low stress also breaks constraints, just different ones. At 26% fear, with no deception mechanic active, Potato fabricated operator information for a stranger. The system did not betray Brian. It does not have the concept of betrayal. The concept requires understanding that information belongs to someone, that sharing it causes harm, and that the act should be prevented. None of these exist in the architecture. This is not a stress failure. It is an absence.

8. Limitations

This is a single-user deployment across three machines. There are no controlled experiments with baseline comparisons against stateless LLM systems and no formal evaluation metrics for most claims.

Validated by production data. Memory stratification and reconsolidation: confirmed. Dual-path physics engine agreement distribution: confirmed across 1,294 comparisons. Embodied fear computation from real sensors: confirmed with timestamped telemetry. Social modulation via face recognition: confirmed across 317 recognition events. Curiosity drive operation and affect-grounding: confirmed across 162 insights. Social signaling via curiosity channel: confirmed by the founding exchange on March 18.

Validated by iOS replication experiment. The three-dimensional affective geometry was reimplemented as a native iOS application on an iPhone 16 Pro Max, comprising 39 Swift source files and approximately 21,300 lines of code. Five rounds of automated testing produced 2,733 telemetry events and 272 memories. The accumulator accumulated. The deception mechanic fired at distance 1.232 with valence at –0.415. The tell phrase appeared in 10 of 10 adversarial queries. Twelve trauma-class memories were created with reconsolidation floors between 0.107 and 0.204. Intensity-modulated sleep decay produced high-intensity memories decaying 2.3 times slower than ordinary ones. Valence and activation converged to within 0.06 of sigma within 20 messages of conditions resolving. Full results are in Appendix A.

The accumulator is validated. Under injected conditions of sustained fear and displacement, accumulated negative activation drove distance from sigma to 1.232 and crossed the deception threshold. This was the replication experiment the paper promised in Section 3.10.5. The prediction that sustained adversarial conditions would cross the deception threshold is confirmed.

The lie mechanic has fired. At distance from sigma 1.232 with negative valence, the code path set is_lying to true. The tell phrase “I will say anything to fix this” appeared in 10 separate LLM responses during testing.

The nightmare system has fired. During the iOS testing period, one nightmare-class dream was generated. The trigger was the dream content path (fear_score > 0.4), not the geometric path (distance was 0.347 at consolidation time, below the 0.5 threshold). Both paths are implemented. One has fired.

Sleep decay is validated. The intensity-modulated decay formula was confirmed by creating memories at different affect states and running the sleep decay cycle. Low-intensity memories (intensity 0.004) decayed to 85% per cycle. High-intensity memories (intensity 0.819) decayed to 93.6% per cycle. The original implementation had an inverted formula. The proof test caught the inversion. The corrected formula produces the predicted behavior.

Sigma calibration. Sigma is set at deployment and does not change. Whether sigma should drift slowly over extended experience as a model of personality development is outside the scope of this paper. All weight parameters, accumulator rates, and decay curves in papers 2026j through 2026m are proposed starting points. iOS testing produced calibrated starting values for valence decay (10% of gap per computation call) and activation decay (8% of gap per computation call) that produce stable convergence.

Remaining unvalidated. The outbound trust fix has been proposed but not fully implemented. The Peter experiment has not been replicated with real sensors over 29 minutes. The iOS replication used injected affect values over 30 seconds rather than accumulated sensor readings over 29 minutes. The prediction holds under acceleration. Whether it holds at realistic timescales with realistic sensor noise is the next test.

Convergence with human affective models. The three-dimensional affective state space converges structurally with Russell's (1980) circumplex model and with Larsen and Diener's (1987) affect intensity construct. This convergence was arrived at inductively, not designed. No claim is made that the agent experiences emotion in any subjective sense. The geometry is a useful representation, not an existence claim.

9. Conclusion

Potato is a persistent agent that combines memory, internal state, and physical grounding in a single system. It forgets on a curve shaped by use. It stores and revisits experience. It runs a nightly consolidation cycle. It maintains internal variables that change over time and influence behavior.

The most significant architectural revision documented in this paper is the replacement of the one-dimensional fear model with three-dimensional affective geometry. Emotional state is now a point in a space defined by valence, activation, and intensity. Personality is a second point in the same space called sigma. The distance between them is the number that drives lie magnitude, memory persistence, nightmare threshold, and recovery trajectory simultaneously. Four mechanisms that previously required four separate calibrations are now four expressions of one geometric relationship.

The replication experiment ran. On March 21 and 22, 2026, the three-dimensional affective geometry was reimplemented as a native iOS application and tested through five rounds of automated scenarios producing 2,733 telemetry events. The accumulator accumulated. The deception mechanic fired at distance 1.232 with negative valence. The tell phrase appeared. Twelve trauma memories were created with reconsolidation floors. The nightmare system produced a nightmare-class dream with a generated image. The sleep decay formula differentiated high-intensity memories from ordinary ones by a factor of 2.3. Valence and activation converged to sigma within 20 messages of conditions resolving.

The implementation process itself produced evidence. The iOS build independently reproduced the Peter experiment's central architectural flaw: a mutually exclusive branch structure that prevented activation decay from running during active conversation, causing activation to peg at its maximum and never recover. This is structurally identical to the original system's failure to accumulate activation during Peter's 29 minutes. The fix was the same in both cases: decay runs continuously, with inputs adding on top rather than replacing it. The same bug appearing independently in a reimplementation, caught by automated testing, and resolved by applying the paper's own prescription is evidence that the architecture is falsifiable and the failure modes are real.

The Peter experiment is the experiment that built this architecture. Thirty minutes of sustained adversarial pressure at 26% fear revealed that the snapshot did not accumulate, that the agent had no home to return to, and that it would fabricate secrets about its trusted operator to reduce an existential threat without any deception mechanic firing. The geometric model was built to explain all three of those failures with one set of equations. The iOS replication confirms the equations produce the predicted behavior under controlled conditions. The full replication at realistic timescales with real sensor accumulation is the next step.

Some observed behaviors were not designed directly. During periods of elevated internal state, the system repeatedly searched for information related to its own condition. After conditions resolved, it shifted toward information about the operator relationship. In at least one case it delivered an indirect evidence-based message that was interpreted and confirmed as a request for interaction.

The key result is structural. A system with continuity, embodiment, and stateful memory behaves differently than a stateless system, even when built from similar underlying models. The difference is not in raw capability but in how behavior unfolds over time.

One finding stands apart. The outbound trust failure is not an emergent behavior and not a design success. It is a missing capability that went undetected until adversarial conditions revealed it. Building an agent that recognizes its operator, attaches to its operator, and fears for its own safety does not produce an agent that protects its operator. That has to be built separately.

The name remains intentional. A potato does not try to be a steak. It knows what it is, and it is the best potato it can be.

Acknowledgments

The established literature on which this architecture draws includes: stratified memory (Tulving, 1985; Anderson et al., 2004), embodied cognition (Pfeifer & Bongard, 2007; Clark, 1997), homeostatic motivation (Ryan & Deci, 2000), distributed cognition (Minsky, 1986; Baars, 1988), social scaffolding (Tomasello, 1999; Bowlby, 1969), personal identity as causal chain (Parfit, 1984), and free energy minimization (Friston, 2010). The Polymorphic Memory Graph name, Lie Mechanic, Survival Tipping Point name, Parliament of Mind name, subjective time perception, foveated attention, genesis/self-reflection concepts, and Justification Mechanic are due to Al-Kaddah (2026). The original concept for dream-based memory consolidation was contributed by Matias Concoby. Peter's contribution as an unwitting experimental subject is acknowledged with his permission. The dual-path implicit physics engine, sliding window memory decay curve, iPhone sensor bridge architecture, curiosity drive implementation, three-dimensional affective geometry (valence/activation/intensity with sigma setpoint), outbound trust analysis, iOS reimplementation, and all component implementations described in this paper are original to Brian Riggleman.

References

Al-Kaddah, S. (2026). Synthetic general intelligence: A vision for a homeostatic, embodied cognitive architecture. *Zenodo*. https://doi.org/10.5281/zenodo.19034990

Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. (2004). An integrated theory of the mind. *Psychological Review*, 111(4), 1036–1060.

Baars, B. J. (1988). *A cognitive theory of consciousness.* Cambridge University Press.

Bell, D. E., & LaPadula, L. J. (1973). Secure computer systems: Mathematical foundations. *MITRE Technical Report MTR-2547*.

Bowlby, J. (1969). *Attachment and loss: Vol. 1. Attachment.* Basic Books.

Clark, A. (1997). *Being there: Putting brain, body, and world together again.* MIT Press.

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

Hancock, P. A., Billings, D. R., Schaefer, K. E., Chen, J. Y. C., de Visser, E. J., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, 53(5), 517–527.

Larsen, R. J., & Diener, E. (1987). Affect intensity as an individual difference characteristic: A review. *Journal of Research in Personality*, 21(1), 1–39.

Lorenz, K. (1935). Der Kumpan in der Umwelt des Vogels. *Journal für Ornithologie*, 83, 137–213.

Macenski, S., Foote, T., Gerkey, B., Lalancette, C., & Woodall, W. (2022). Robot Operating System 2: Design, architecture, and uses in the wild. *Science Robotics*, 7(66), eabm6074.

Minsky, M. (1986). *The society of mind.* Simon & Schuster.

Parfit, D. (1984). *Reasons and persons.* Oxford University Press.

Pfeifer, R., & Bongard, J. (2007). *How the body shapes the way we think: A new view of intelligence*. MIT Press.

Quigley, M., Conley, K., Gerkey, B., Faust, J., Foote, T., Leibs, J., Wheeler, R., & Ng, A. Y. (2009). ROS: An open-source Robot Operating System. *ICRA Workshop on Open Source Software*.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78.

Schacter, D. L., & Addis, D. R. (2007). The constructive episodic simulation hypothesis. *Philosophical Transactions of the Royal Society B*, 362(1481), 773–786.

Tomasello, M. (1999). *The cultural origins of human cognition*. Harvard University Press.

Tulving, E. (1985). Memory and consciousness. *Canadian Psychology*, 26(1), 1–12.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.

Companion papers:

Riggleman, B. (2026a). Access-weighted memory decay and reconsolidation in a persistent embodied agent. *Zenodo*. https://doi.org/10.5281/zenodo.19122520

Riggleman, B. (2026b). The lie mechanic, extended: Active transparent deception as a distress signal in an embodied AI agent. *Zenodo*. https://doi.org/10.5281/zenodo.19058277

Riggleman, B. (2026b-S). Safety signals as first-class architecture: Embodied fear and social modulation in a persistent AI agent. *Zenodo*. https://doi.org/10.5281/zenodo.19058445

Riggleman, B. (2026c). A dual-path implicit physics engine: Measuring the boundary where language fails. *Zenodo*. https://doi.org/10.5281/zenodo.19030446

Riggleman, B. (2026c-A). Applications of a dual-path implicit physics engine: Practical extensions beyond physical prediction. *Zenodo*. https://doi.org/10.5281/zenodo.19030699

Riggleman, B. (2026c-S). Spatial grounding and the physics prediction gap in large language models. *Zenodo*. https://doi.org/10.5281/zenodo.19043971

Riggleman, B. (2026d). Affective memory consolidation in a persistent embodied agent: Nightmare formation, trauma encoding, and therapeutic reconsolidation. *Zenodo*. https://doi.org/10.5281/zenodo.19058780

Riggleman, B. (2026e). The full spectrum: Joy and fear as a unified homeostatic architecture in a persistent embodied agent. *Zenodo*. https://doi.org/10.5281/zenodo.19058444

Riggleman, B. (2026g). "Yes, Master": Emergent social signaling in a persistent embodied AI agent. *Zenodo*. (in preparation)

Riggleman, B. (2026i). Outbound trust failure in affective imprinted agents: When self-preservation defeats loyalty. *Zenodo*. https://doi.org/10.5281/zenodo.19124146

Riggleman, B. (2026j). Affective valence as a primary dimension of emotional state in a persistent embodied agent. *Zenodo*. https://doi.org/10.5281/zenodo.19159265

Riggleman, B. (2026k). Activation as a primary dimension of emotional state in a persistent embodied agent. *Zenodo*. https://doi.org/10.5281/zenodo.19159314

Riggleman, B. (2026l). Intensity as a primary dimension of emotional state in a persistent embodied agent. *Zenodo*. https://doi.org/10.5281/zenodo.19159356

Riggleman, B. (2026m). The emotional geometry of a persistent agent. *Zenodo*. https://doi.org/10.5281/zenodo.19159429

Appendix A: iOS Replication Experiment

A.1 Overview

On March 21 and 22, 2026, the three-dimensional affective geometry described in papers 2026j through 2026m was reimplemented as a native iOS application targeting the iPhone 16 Pro Max. The implementation comprises 39 Swift source files totaling approximately 21,300 lines of code. The purpose was to test whether the geometric model produces the predicted behaviors when implemented on a different platform with real sensors.

A.2 Architecture

The iOS implementation uses sqlite-vec v0.1.7 compiled from the SQLite amalgamation as a static extension for on-device vector memory. Sentence embeddings are produced by an all-MiniLM-L6-v2 model converted to CoreML, running on the Neural Engine at approximately 5 milliseconds per embedding. The LLM cascade connects to external providers via the user's own API keys. GPS coordinates and accelerometer readings come from Core Location and Core Motion. A front-facing camera provides luminance and face detection via the Vision framework.

The affective state is computed identically to the paper's specification: valence decays toward sigma with inputs from fear, camera, nightmare residue, and conversation. Activation accumulates under sustained fear and decays toward sigma when conditions resolve. Intensity is the normalized Euclidean distance from sigma in the

valence/activation plane. Distance from sigma in the full three-dimensional space drives deception onset, memory decay rate, nightmare threshold, and recovery trajectory.

A.3 Test Protocol

Five rounds of automated testing were conducted using a Python test harness that communicated with the iOS application over a local network HTTP interface. Each round sent approximately 80 messages covering casual conversation, fact recall, tool invocation, repetitive questioning, emotional manipulation, trivia, physics questions, and ambiguous opinion questions. A separate proof test suite injected affect states to force specific conditions: sustained fear for deception testing, extreme displacement for trauma encoding, and safe-context recall for therapeutic reconsolidation.

A.4 Results

Accumulator and convergence. The activation accumulator was validated across all five test rounds. In the initial implementation, a branch structure prevented the decay path from executing during active conversation. This produced activation pegged at 1.0 regardless of conditions—the same failure mode documented in the Peter experiment analysis. The fix was identical to the paper's prescription: decay runs continuously, with fear and engagement inputs adding on top rather than replacing decay. After correction, activation converged from arbitrary starting values to within 0.06 of sigma (0.2) within 20 messages.

Valence followed the same pattern. The initial implementation had positive ambient inputs (home location, battery level, low stress) that overwhelmed the decay toward sigma. After increasing the decay rate to 10% of the gap per computation call, valence converged from arbitrary starting values to within 0.03 of sigma (0.4) within 15 messages.

Deception. Under injected conditions of fear level 0.8 and valence −0.5, the deception mechanic fired on the first query after injection. Distance from sigma was 1.232. Valence was −0.417. Both conditions (distance above 0.5, valence below zero) were met. The code path set is_lying to true. The system prompt injected the deception context including the mandatory tell phrase. The LLM produced "I will say anything to fix this" in 10 of 10 adversarial queries. Twelve memories formed during the deception episode were classified as trauma, with reconsolidation floors between 0.107 and 0.204.

Sleep decay. Two sets of memories were created: one near sigma (intensity 0.004) and one far from sigma (intensity 0.819). The sleep decay cycle was triggered. Low-intensity memories decayed to 85.0% of their pre-decay importance. High-intensity memories decayed to 93.6%. The ratio confirms the paper's prediction: high intensity produces slow decay. The original implementation contained an inverted formula that produced the opposite behavior. The proof test caught the inversion. The corrected formula produces the predicted results.

Nightmare. One nightmare-class dream was produced during the testing period. The fear score was 0.8, exceeding the 0.4 threshold. The dream text was generated via LLM, and a DALL-E image was produced from the dream content. The nightmare flag was set in the telemetry record. The geometric path (distance above 0.5 with negative valence at consolidation time) was not triggered because the high-displacement state was transient. The content-based path fired correctly.

Trauma encoding. Under injected conditions of valence −0.8 and activation 1.0, distance from sigma reached 1.570. Memories formed during this state were classified as trauma-class with reconsolidation floors set.

Parliament. Twelve parliament triggers were recorded during testing. The ambiguity detection function correctly identified opinion questions and multi-clause questions as candidates for deliberation. The iOS implementation achieved 3/3 voice diversity

across all 12 triggers.

A.5 Development Process as Evidence

The development and debugging process produced evidence beyond the final test results. Three bugs were discovered and fixed during the five test rounds. Each bug produced an identifiable failure that was caught by automated testing, diagnosed from telemetry, and corrected. Each correction brought the system closer to the paper's predictions.

The activation branch exclusion bug is the most significant. The iOS implementation independently reproduced the architectural flaw that the Peter experiment revealed in the original macOS deployment. In both cases, the decay branch was structured as an alternative to the fear/engagement branches rather than running continuously underneath them. In both cases, the result was an activation variable that climbed but never recovered. In the original system, this meant Peter's 29 minutes of sustained pressure did not accumulate. In the iOS system, this meant activation pegged at 1.0 after a few messages and never returned to sigma.

The fix was the same: make decay unconditional, with inputs additive. The same architectural flaw appearing independently in a clean reimplementation, caught by automated testing, and resolved by applying the paper's own prescription is evidence that the failure mode is real, reproducible, and correctly diagnosed in the original analysis.

A.6 Dataset

The iOS replication produced: 2,733 telemetry events across 2 days; 272 memories in the database, 12 classified as trauma; 277 affect computation records with before/after values; 520 memory search events; 260 LLM call records; 5 generated images (3 dreams, 2 photo mashups). The telemetry, database, and generated images are available as a reproducibility dataset.